

BHANU NAGPURE

Software Engineer

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SUMMARY

Software engineer (B.Tech CSE, CGPA 8.7) who owns the whole product lifecycle - architecture, backend, and shipping. Shipped **3 apps to Google Play (2 live, 1 in closed testing)** with offline-first sync, event-driven async engines, REST/Firebase backends and CI/CD pipelines. Globalink intern (Mitacs, Canada); **IGI Global** and **ACM** publications; **granted patent**; Wikimedia OSS.

TECHNICAL SKILLS

Languages: Java (primary), Python, Kotlin, C++, SQL, JavaScript

Data Structures & Algorithms: Arrays, Linked Lists, Trees, Graphs, Hash Maps, Heaps, Dynamic Programming, Sorting & Searching, complexity analysis

Backend & Systems: REST APIs, Firebase (Firestore, Auth, Storage, Crashlytics), SQL & NoSQL, asynchronous pipelines, offline-first sync, event-driven architecture

Concurrency & Architecture: Java threading, Kotlin Coroutines & Flow, WorkManager; Clean Architecture, MVVM/MVI, SOLID, Hilt, Repository pattern

DevOps & Cloud: Git, GitHub Actions CI/CD, Gradle, Google Cloud Platform, Firebase, Docker, Linux

PROFESSIONAL EXPERIENCE

Devanshu Studios - Founder & Software Engineer

Aug 2025 - Present

Independent - Remote

- Took complete ownership of building and shipping **PyMaster** (Play Store): 50k+ LOC production platform, **Clean Architecture**, multi-env build system (dev/staging/prod), Riverpod 2.0 and a Gemini AI tutor - **99.5% crash-free sessions**; designed unit-testable data / domain / presentation layers with Repository + Hilt DI, Proto DataStore and EncryptedSharedPreferences.
- Engineered an **offline-first async sync engine**: event queue persists user progress locally and batch-uploads to Firebase on reconnect - **zero data loss** under intermittent connectivity with deterministic conflict resolution.
- Owned the full **production lifecycle**: CI/CD pipeline, Play Console publishing, A/B testing, Crashlytics incident response, RevenueCat billing (subscriptions + IAP) and AdMob native-ad mediation; shipped **Chakuli** (live: Gemma 4, Qwen3.5, Phi, SmolLM on a dual **LiteRT + GGUF** runtime) and led **Keepary** through Play closed testing (on-device ML Kit OCR, RevenueCat Pro).

Athabasca University - Globalink Research Intern (Mitacs)

Apr 2024 - Oct 2024

Machine Learning & Software Development - Edmonton, AB, Canada

- Built a text-ideology classifier in Java/Python using **BERT/Transformers** reaching **95% prediction accuracy**; deployed as a REST-backed web service for researcher validation.
- Sole developer of a cross-platform (iOS, Android, Web) field-trip app - Firebase Auth/Storage backend, OpenStreetMaps integration; presented at the **TRESL Lab International Symposium**, Canada (Jul 2024).

BETIC-GHRCE - Software Developer Intern

Oct 2022 - May 2024

Nagpur, India

- Built a production Java/Kotlin diagnostic app with MediaPipe FaceMesh - **60%** accuracy improvement and **30%** inference speedup via camera-pipeline optimisation; contributed to a **granted Indian patent** (TMJ Screening Tool).

Wikimedia Foundation - Open Source Contributor

Feb 2025 - Present

Scribe-Android (Kotlin) - Remote

- Implemented tablet keyboard layouts for 7+ languages; refactored the core **KeyHandler** module to reduce coupling and improve testability; resolved critical UI bugs.

KEY PROJECTS

PyMaster - Learn Python | Dart, Kotlin, Firebase, Riverpod 2.0, Gemini AI, RevenueCat

2025 - Present

- **Live on Google Play**. 1,200+ exercises, gamified leagues, Gemini AI tutor and offline-first sync - 50k+ LOC at 99.5% crash-free. Multi-env Flavors, RevenueCat IAP, AdMob mediation.

Chakuli - Offline AI Chatbot | Kotlin, LiteRT-LM + GGUF, Jetpack Compose, WorkManager, Proto DataStore

2025 - Present

- **Live on Google Play**. Runs Gemma 4, Qwen3.5, Phi 3.5 Mini and SmolLM entirely on-device on a dual LiteRT + GGUF runtime - zero cloud inference. Compose UI, Hilt DI, CameraX, Agent Skills, encrypted key storage.

Rote Playoffs - Agent Procedure Benchmark (WeMakeDevs x Modiqo) | Python, Rote CLI, controlled A/B benchmarking, deterministic grading

Sep 2026

- Shipped **38 agent procedures** (all 1.00, top-5 of 123 makers); measured procedure reuse in a controlled study: **9.9x faster**, **15.3x fewer output tokens**, replicated across 6 tasks with frozen ground truth.

EDUCATION

G H Raisoni College of Engineering

Dec 2021 - Jun 2025

B.Tech in Computer Science & Engineering | CGPA: 8.7 / 10.0

Nagpur, India

Operating Systems, Computer Networks, DBMS, OOP, Data Structures & Algorithms

PUBLICATIONS, PATENTS & AWARDS

- **Journal**: "V-Safe-Anywhere: Empowering Women's Safety with Wearable AI and IoT." *IGI Global*, Mar 2024.
- **Conference**: "Autonomous Floating Garbage Collection Device using Computer Vision and Robotics." *8th ICACS 2024, ACM, Hong Kong*, Oct 2024.
- **Conference**: "Development of a Mobile Field Trip Application." *3rd Annual TRESL Lab International Symposium*, Athabasca, Canada, Jul 2024.
- **Patent (Granted)**: "Smart Chairside TMJ Examination & Muscle Activity Detector Tool," India (Pub. No. 202321022476).; **Patent (Pending)**: "AI-Based Portable Device for Dental Disease Detection," DPIIT, India (App. No. 202421006731, Published Aug 2025).
- **Awards**: 3rd Place · Avishkar 2024 | Runner-up · DIPEX 2023 & SHRISTI 2023 | BIREC E-YUVA Grant · Rs. 10 lakh seed funding | Winner · MEDHA 2022 & MEDIC-2022